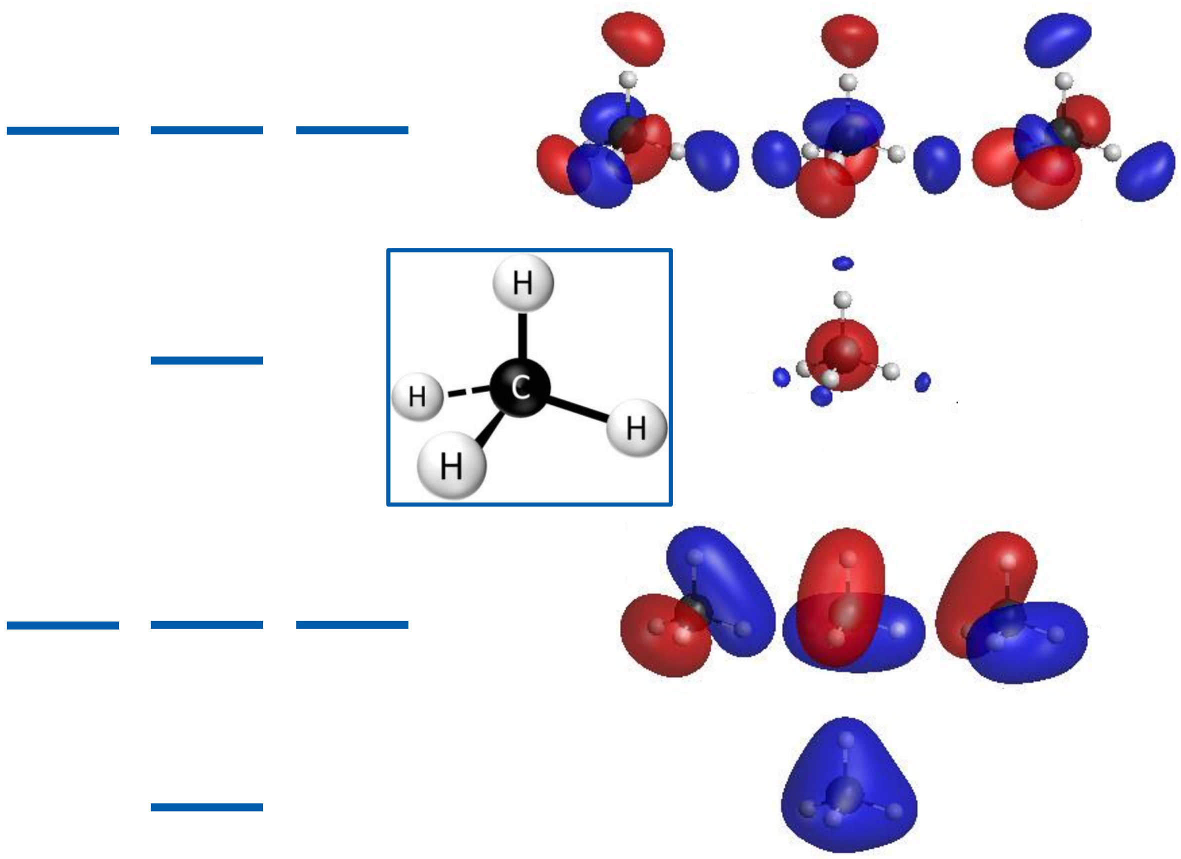
Physical Chemistry (II)

Lecture 23

Polyatomic Molecules (2)

Last class

MOs of Polyatomic Molecules



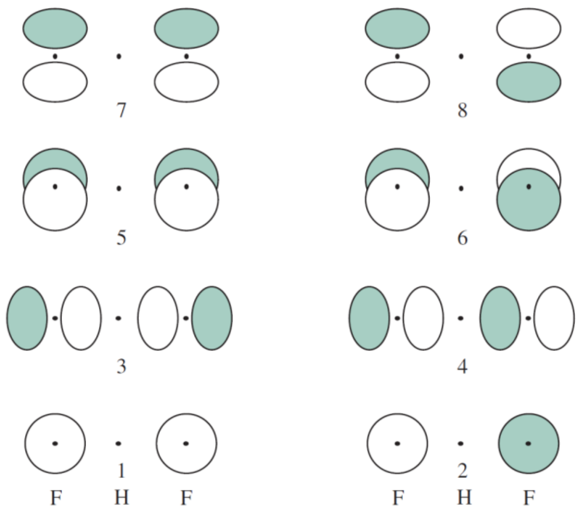
(Image: https://www.flickr.com/photos/69057297@N04/48054283727)

Last class

Ingredients to Construct MOs

- Center atom: AOs
- Outer atoms: combinations of AOs = group orbitals

e.g. FHF^- (bifluoride anion)



(Image: Miessler *et al*., ***Inorganic Chemistry***, 5th Ed., Figure 5.15)

Last class

Point Symmetry Operators

a class of symmetry operators

that leave at least one point unmoved

(the unmoved point = origin)

Identity operator

Inversion operator

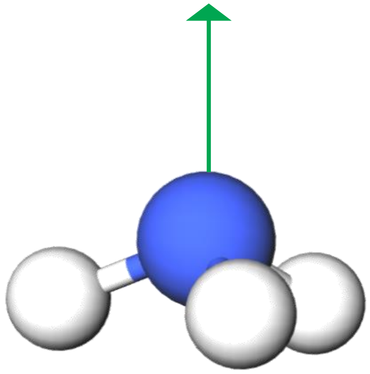
Reflection operator

Rotation operator

Point Group

- A group formed by a set of point symmetry operators belonging to a molecule

- e.g. NH_3 – $\{\hat{E}, \hat{C}_3, \hat{C}_3^2, \hat{\sigma}_a, \hat{\sigma}_b, \hat{\sigma}_c\}$



Character Table

lists the different symmetry types (“irreps”)

possible in a point group

- Character χ
  - shows how an object or mathematical function is affected by the corresponding symmetry operation of the group
  - 1: no change
  - -1: change of sign
  - otherwise: degenerate case (will be explained soon)

Last class

Overlap Integrals and Irreps

- Overlap integral

$$S_{ij} = \int \psi_i^* \psi_j \, \mathrm{d}\tau$$

For nonzero S_{ij}, ψ_i^* and ψ_j must belong to the same irrep

Last class

Projection Operator

$$\hat{P}_j = \sum_{\hat{R}} \chi_j(\hat{R})\hat{R}$$

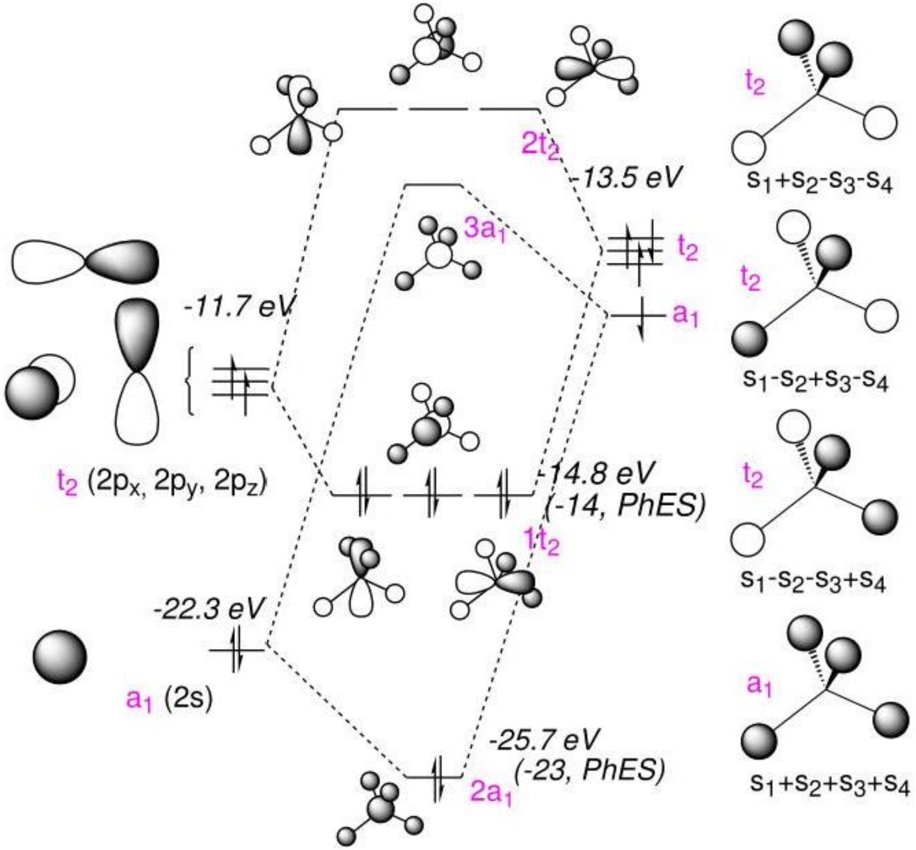
- Useful to find group orbitals belonging to irrep j
- The group orbitals are made of symmetry-adapted linear combinations (SALCs) of atomic orbitals

Today's Topics

1. MO vs. VB theories (example: CH_4): pp. 10-15
2. Potential energy surface (example: H_2O): pp. 16-18
3. Conjugated systems (example: C_2H_4): pp. 19-27
4. Aromaticity and antiaromaticity: pp. 28-30

Not in text

MO Diagram of CH_4



(Image: https://chemistry.stackexchange.com/questions/58774/what-is-responsible-for-making-the-bonds-identical)

RS 10.5
M 10.8

Valence Bond (VB) Theory

H_2 ground state = two ground-state H atoms

$$\psi_{\mathrm{VB}} = \frac{1}{\sqrt{2(1+S^2)}}\left[\psi_{1s,\mathrm{A}}(1)\psi_{1s,\mathrm{B}}(2) + \psi_{1s,\mathrm{B}}(1)\psi_{1s,\mathrm{A}}(2)\right]$$

Can we treat CH_4 by the VB theory?

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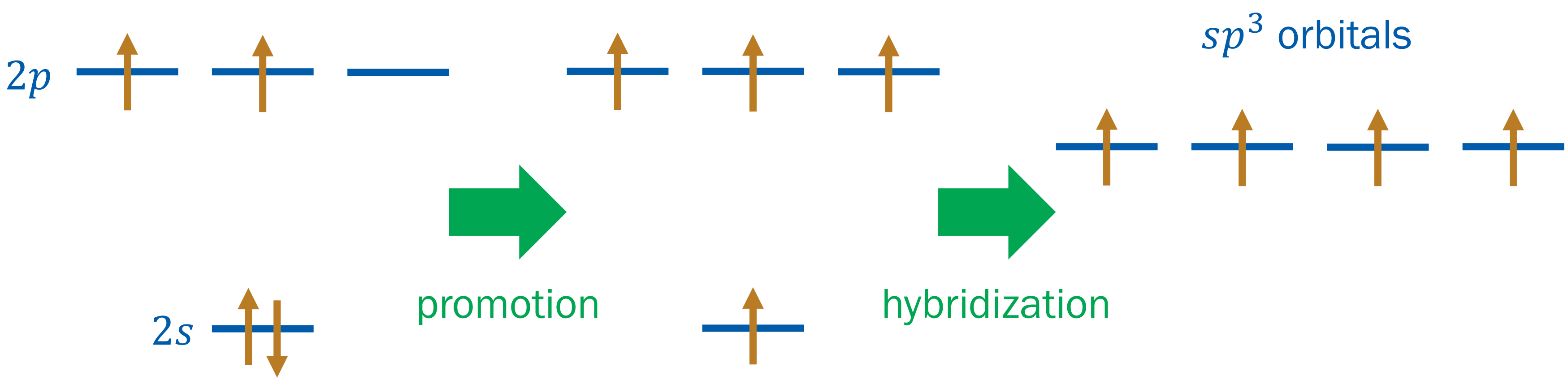
VB Theory: Hybridization

- VB Theory requires singly occupied orbitals to form bonds
- Promotion and hybridization



$$\phi_{sp^3} = c_1\psi_{2s} + c_2\psi_{2p_x} + c_3\psi_{2p_y} + c_4\psi_{2p_z}$$

Not in text

Hybrid Orbitals

$$\phi_{sp^3} = c_1\psi_{2s} + c_2\psi_{2p_x} + c_3\psi_{2p_y} + c_4\psi_{2p_z}$$

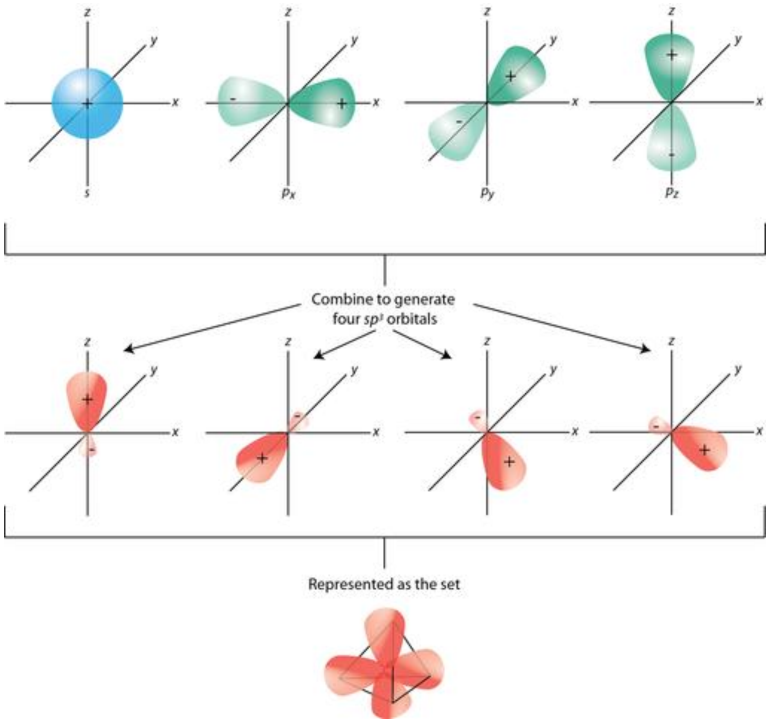
Minimization leads to

$$\phi_{sp^3,1} = \frac{1}{2}\left(\psi_{2s} + \psi_{2p_x} + \psi_{2p_y} + \psi_{2p_z}\right)$$

$$\phi_{sp^3,2} = \frac{1}{2}\left(\psi_{2s} - \psi_{2p_x} - \psi_{2p_y} + \psi_{2p_z}\right)$$

$$\phi_{sp^3,3} = \frac{1}{2}\left(\psi_{2s} + \psi_{2p_x} - \psi_{2p_y} - \psi_{2p_z}\right)$$

$$\phi_{sp^3,4} = \frac{1}{2}\left(\psi_{2s} - \psi_{2p_x} + \psi_{2p_y} - \psi_{2p_z}\right)$$



(Image: https://courses.lumenlearning.com/cheminter/chapter/hybrid-orbitals/)

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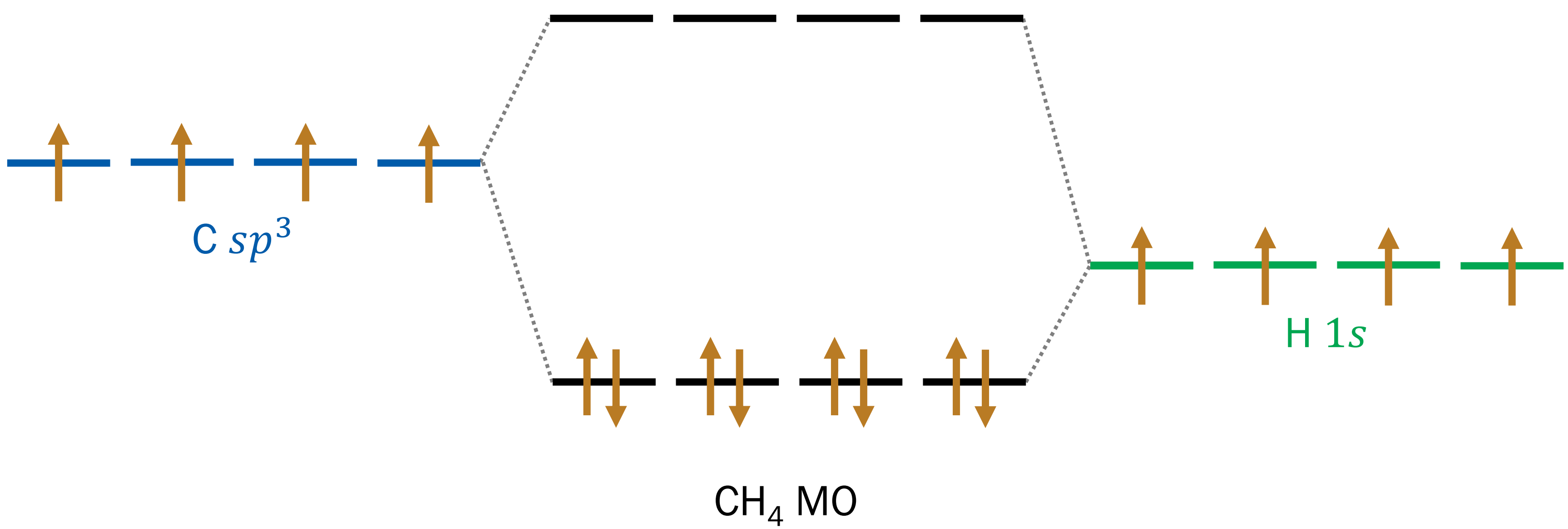
Construction of MO



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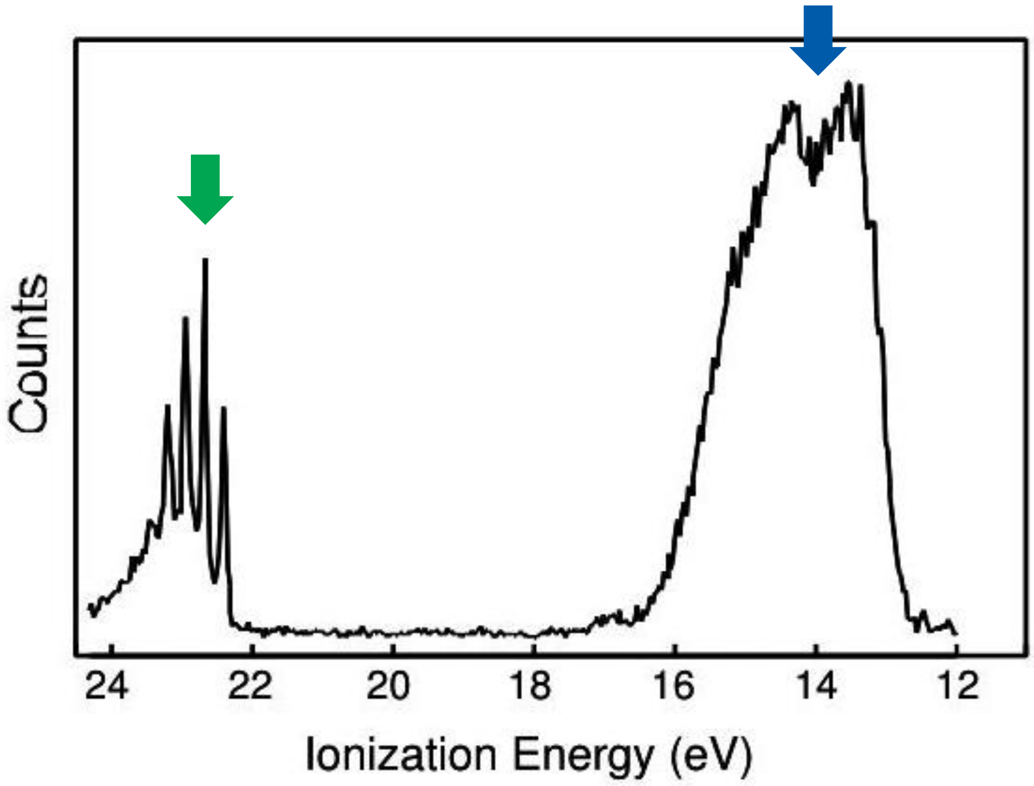
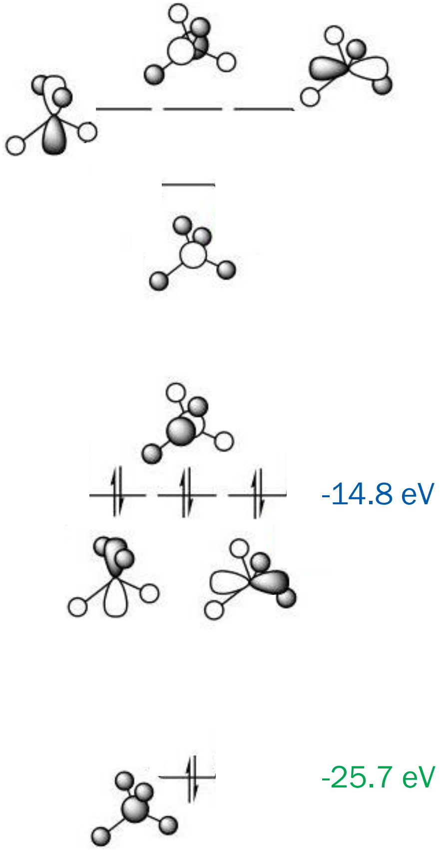
Photoelectron Spectrum of CH_4





(Image: https://www.slideserve.com/kamal-wiggins/photoelectron-spectroscopy)

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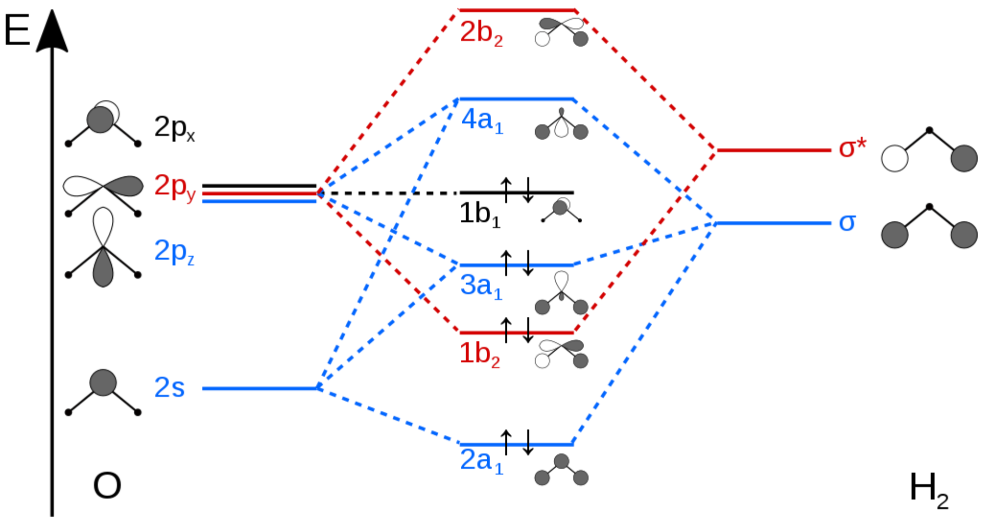
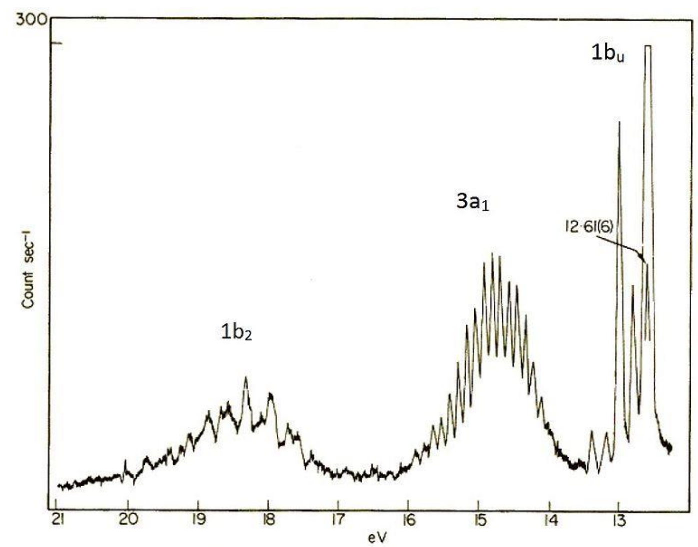
MO Diagram of H_2O





(Images: https://commons.wikimedia.org/wiki/File:H2O-MO-Diagram.svg
https://commons.wikimedia.org/wiki/File:PES_of_H2O.jpg)

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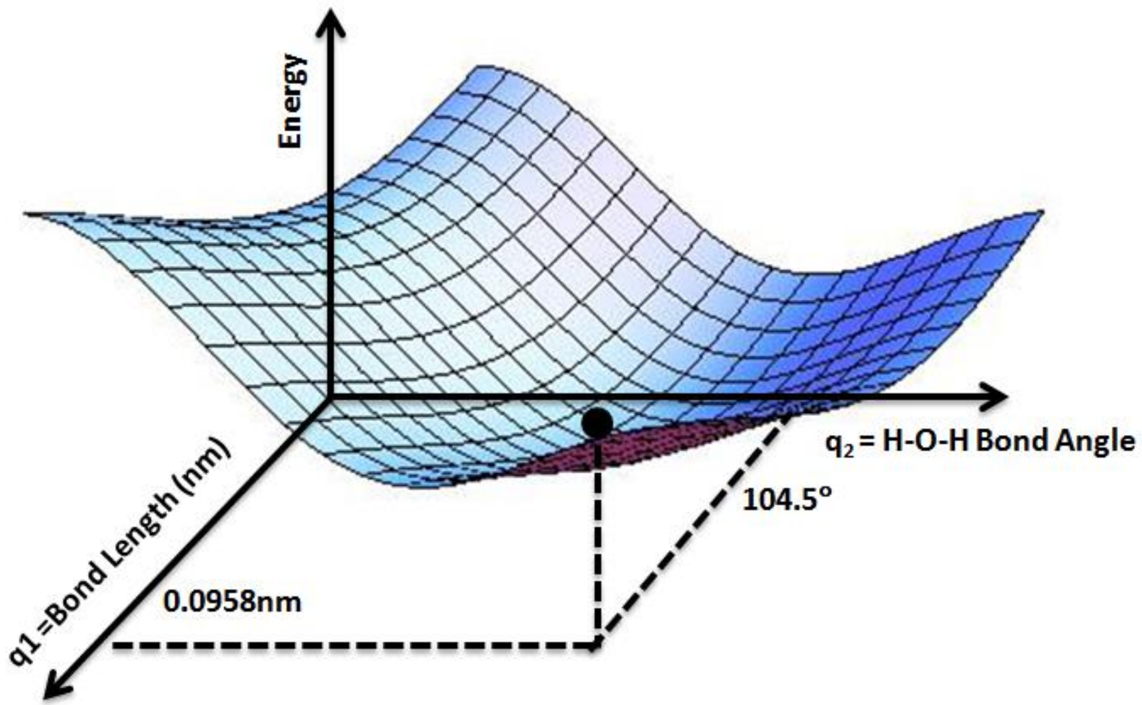
Potential Energy Surface



(Image: https://commons.wikimedia.org/wiki/File:Potential_Energy_Surface_for_Water.png)

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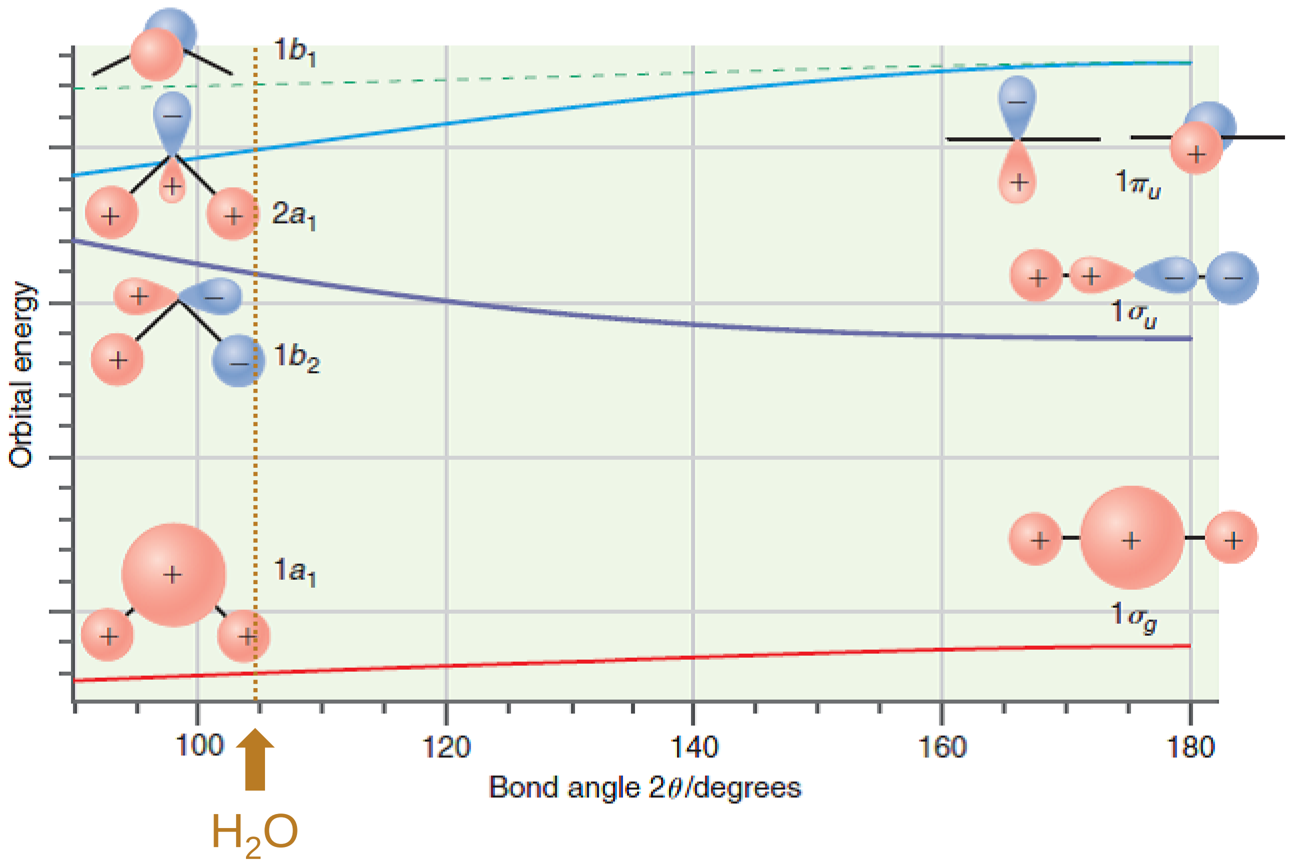
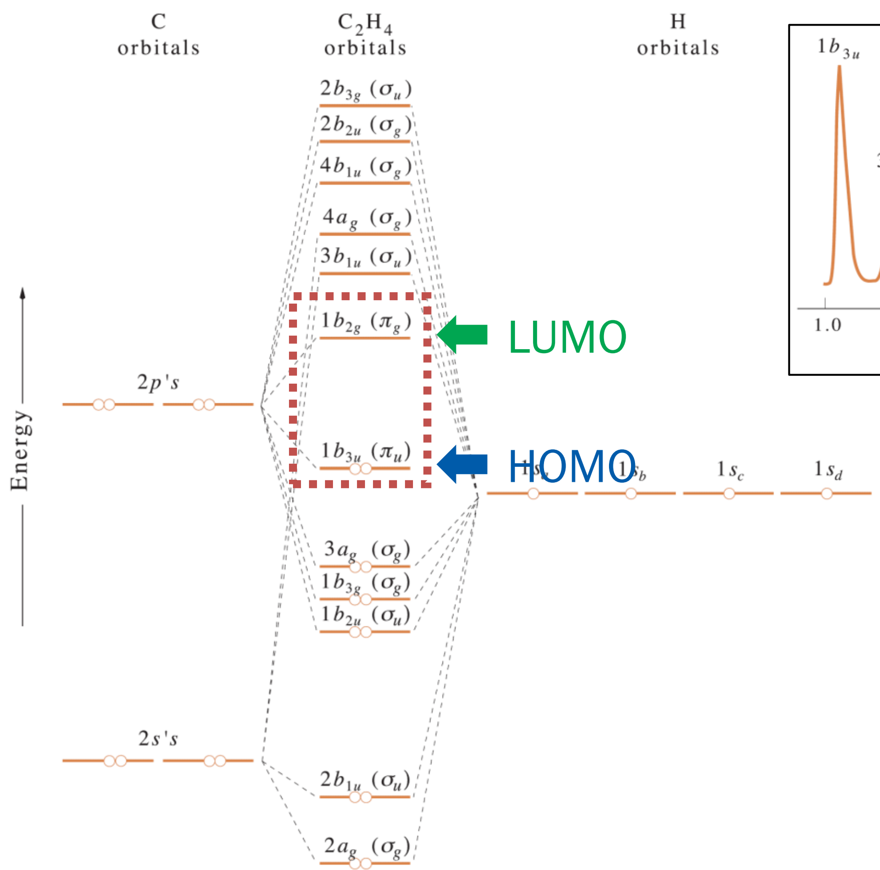
Walsh Diagram: H_2O vs. BeH_2

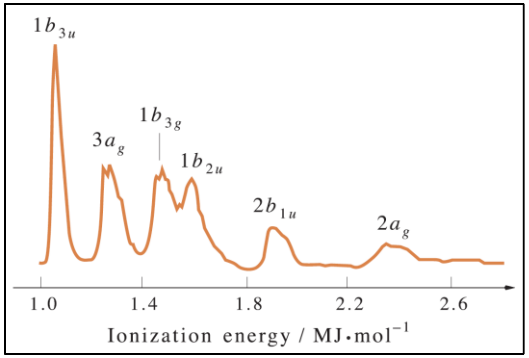


(Image: https://www.chegg.com/homework-help/questions-and-answers/using-walsh-diagram-h2o-assume-bond-angle-90o-write-appropriate-expressions-lcaos-involved-q26081156)

RS 12.1
M 11.6

MO Diagram of Ethene (C_2H_4)



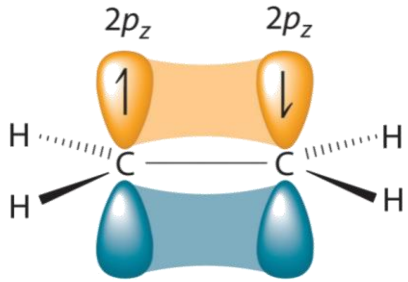


(Image: McQuarrie, ***Quantum Chemistry***, 2^{nd} Ed., Figures 11.13 and 11.14)

RS 12.1
M 11.6

Conjugated Systems

- π-electron approximation: π electrons move in the fixed potential due to the electrons in the σ framework



- Trial function $\phi_\pi = c_1\psi_{2p_z,1} + c_2\psi_{2p_z,2}$
- Minimization leads to a secular equation

$$\begin{vmatrix} H_{11} - ES_{11} & H_{12} - ES_{12} \\ H_{21} - ES_{21} & H_{22} - ES_{22} \end{vmatrix} = 0$$

(Image: https://courses.lumenlearning.com/suny-potsdam-organicchemistry2/chapter/13-1-molecular-orbitals-for-small-%CF%80-systems/)

RS 12.1
M 11.6

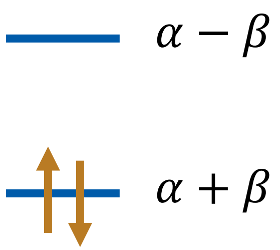
Hückel MO Theory

$$\begin{vmatrix} H_{11} - ES_{11} & H_{12} - ES_{12} \\ H_{21} - ES_{21} & H_{22} - ES_{22} \end{vmatrix} = 0$$

- Simple assumptions
  1. Overlap integral $S_{ii} = 1$ and $S_{ij} = 0$
  2. Coulomb integral $H_{ii} = \alpha$
  3. Exchange integral $H_{ij} = \beta$ for nearest neighbors

$$\begin{vmatrix} \alpha - E & \beta \\ \beta & \alpha - E \end{vmatrix} = 0$$

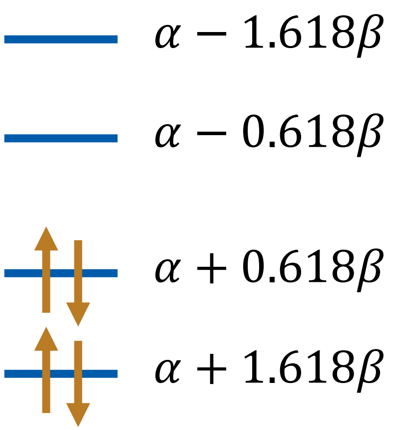
$$E = \alpha \pm \beta$$

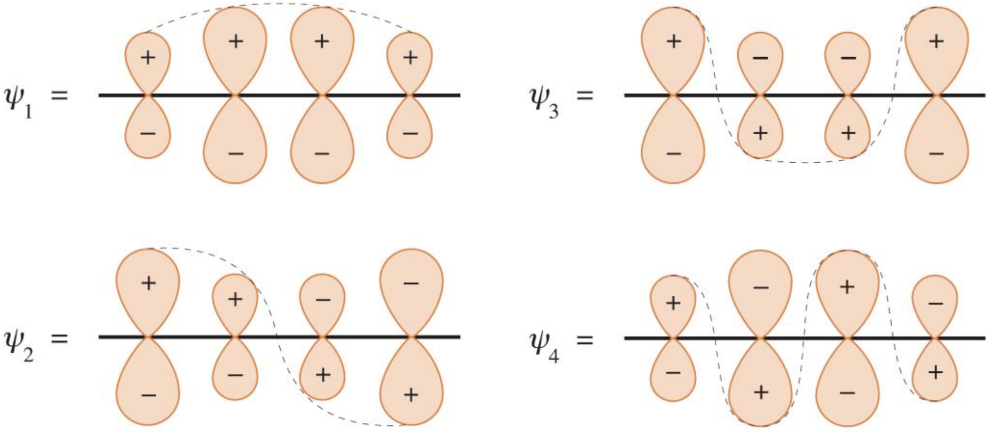


RS 12.1
M 11.6

Butadiene

$$\begin{vmatrix} \alpha - E & \beta & 0 & 0 \\ \beta & \alpha - E & \beta & 0 \\ 0 & \beta & \alpha - E & \beta \\ 0 & 0 & \beta & \alpha - E \end{vmatrix} = 0$$



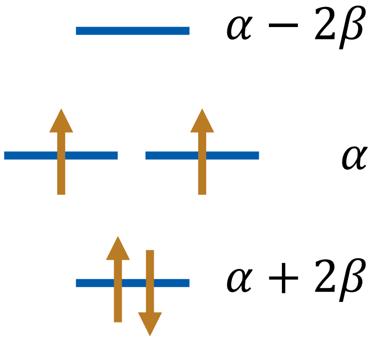


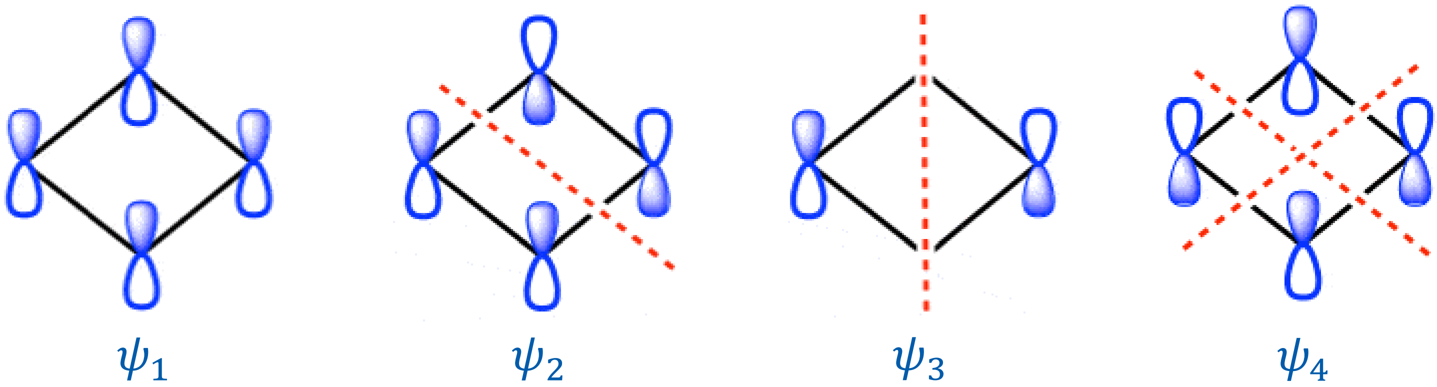
(Image: McQuarrie, ***Quantum Chemistry***, 2nd Ed., Figure 11.19)

RS 12.1
M 11.6

Cyclobutadiene

$$\begin{vmatrix} \alpha - E & \beta & 0 & \beta \\ \beta & \alpha - E & \beta & 0 \\ 0 & \beta & \alpha - E & \beta \\ \beta & 0 & \beta & \alpha - E \end{vmatrix} = 0$$





(Image: https://www.masterorganicchemistry.com/2017/05/10/the-pi-molecular-orbitals-of-cyclobutadiene/)

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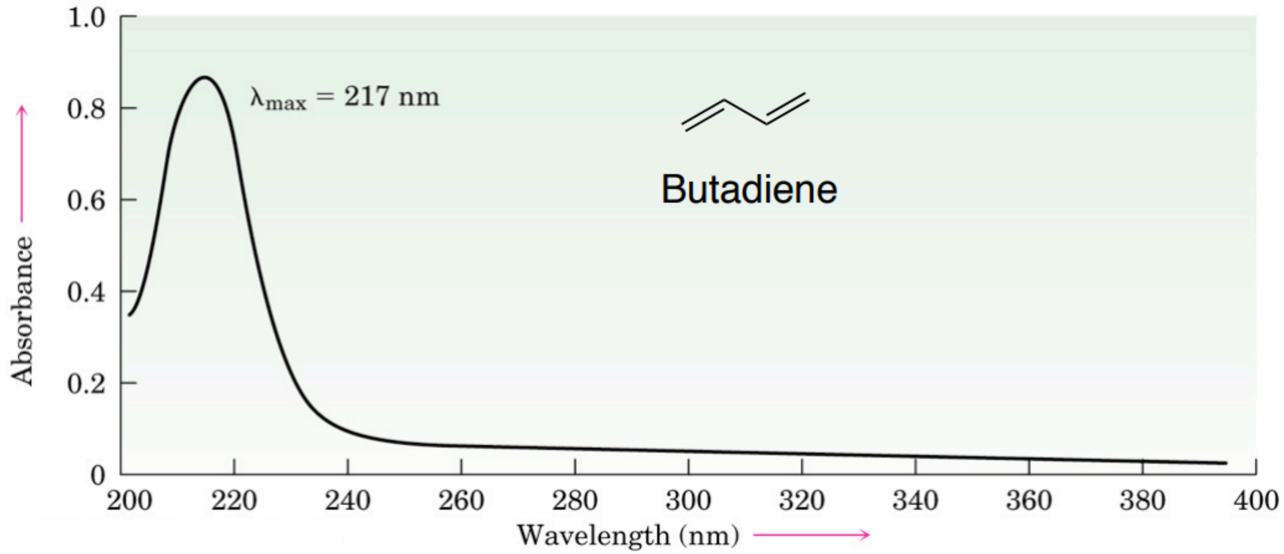
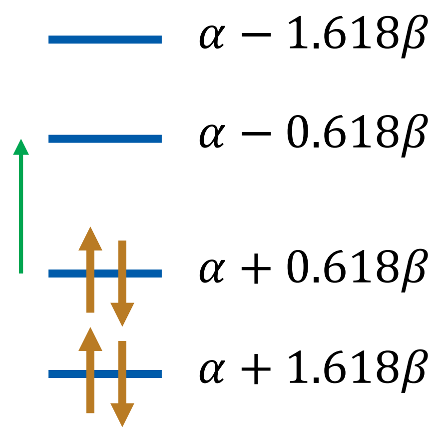
UV-Visible Spectroscopy





(Image: https://www.vanderbilt.edu/AnS/Chemistry/Rizzo/chem220a/Ch14slides.pdf)

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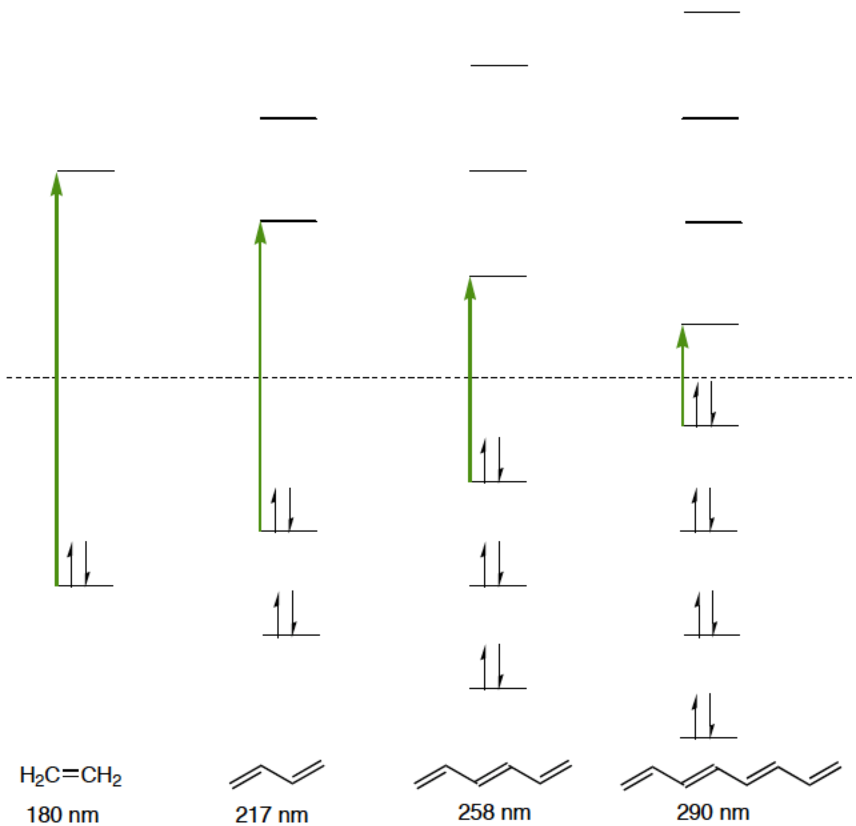
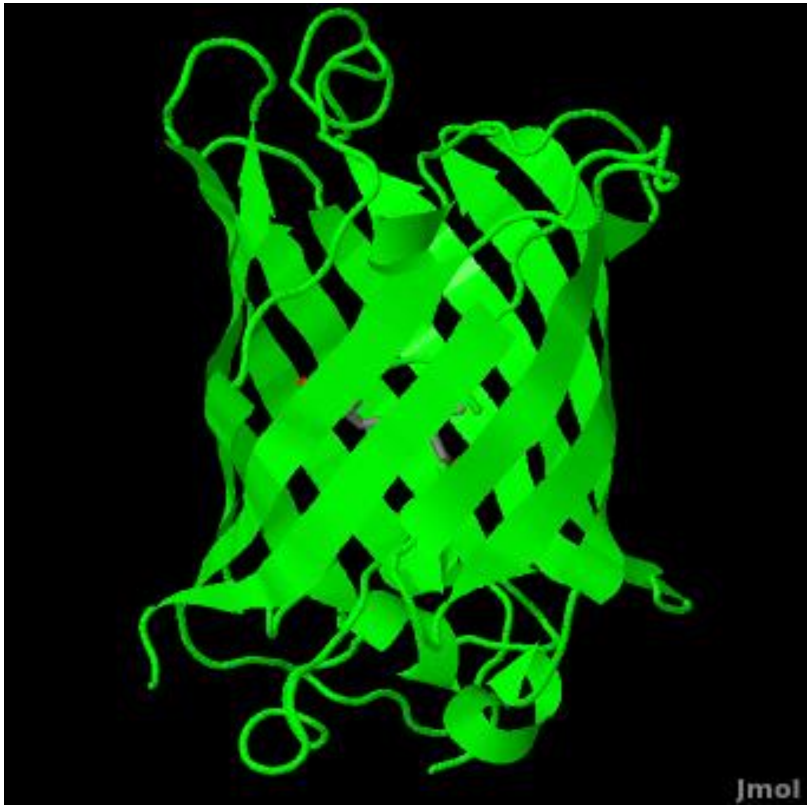
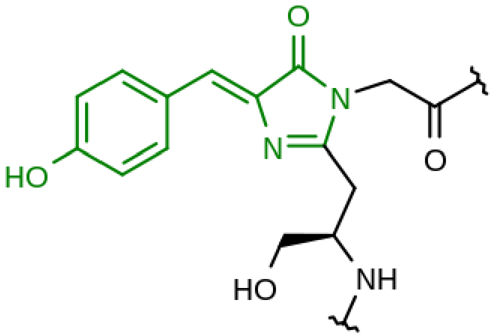
Size of Conjugate System



(Image: https://www.vanderbilt.edu/AnS/Chemistry/Rizzo/chem220a/Ch14slides.pdf)

Green Fluorescent Protein





Nobel prize in Chemistry 2008

(Image: https://proteopedia.org/wiki/index.php/Green_Fluorescent_Protein
https://commons.wikimedia.org/wiki/File:GFP_mechanism.svg)

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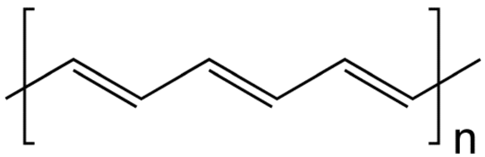
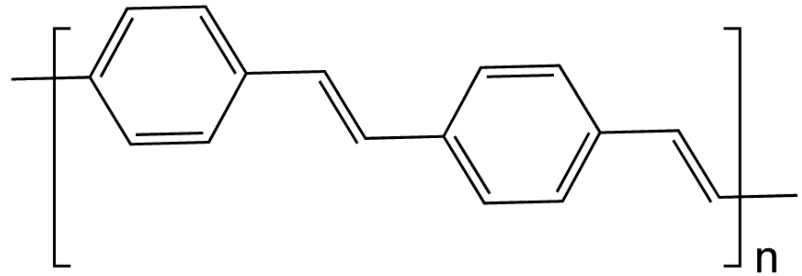
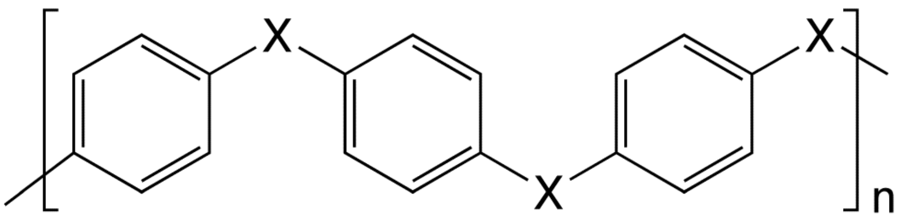
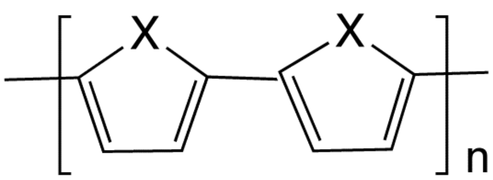
Conductive Polymers







(X = NH/N, S)

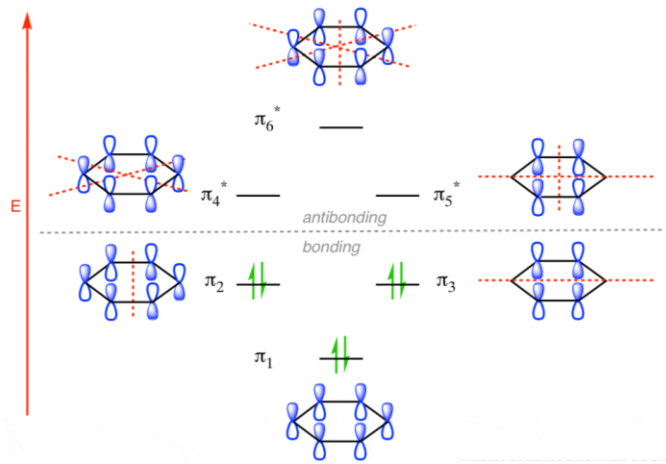


(X = NH, S)

Nobel prize in Chemistry 2000

Benzene

$$\begin{vmatrix} \alpha - E & \beta & 0 & 0 & 0 & \beta \\ \beta & \alpha - E & \beta & 0 & 0 & 0 \\ 0 & \beta & \alpha - E & \beta & 0 & 0 \\ 0 & 0 & \beta & \alpha - E & \beta & 0 \\ 0 & 0 & 0 & \beta & \alpha - E & \beta \\ \beta & 0 & 0 & 0 & \beta & \alpha - E \end{vmatrix} = 0$$



(Image: https://www.masterorganicchemistry.com/2017/05/05/the-pi-molecular-orbitals-of-benzene/)

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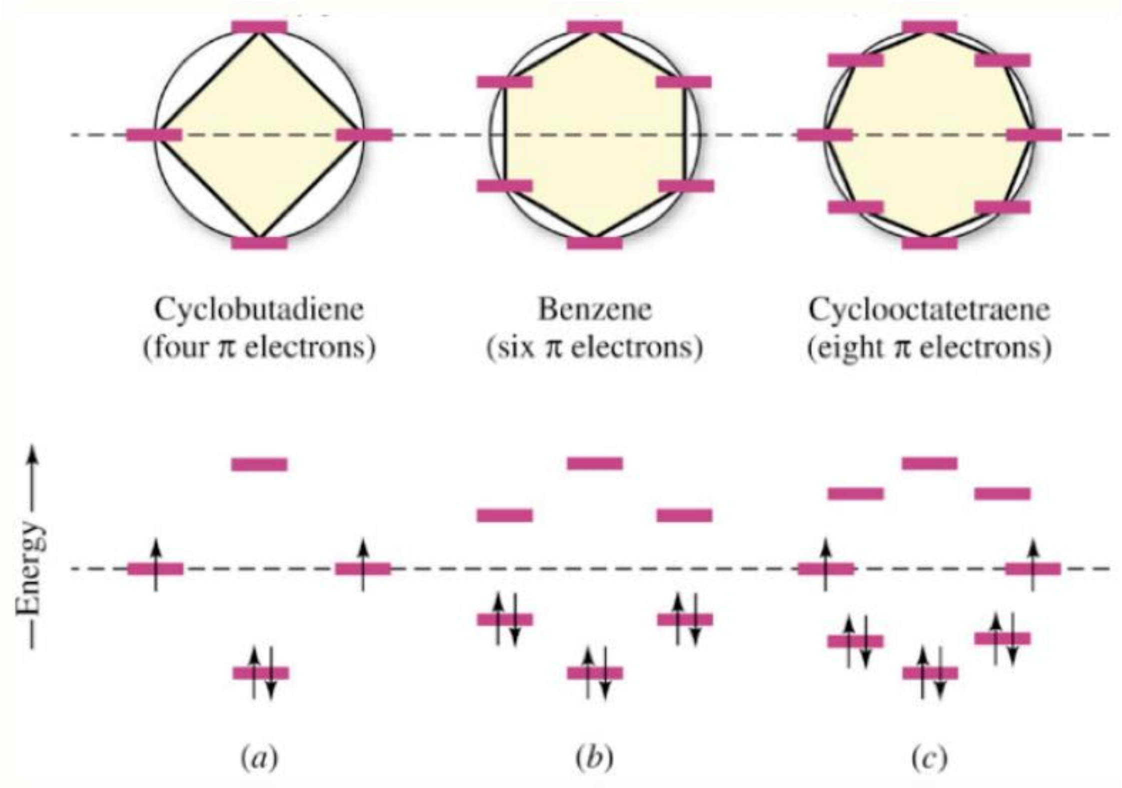
Aromaticity & Antiaromaticity



(Image: http://ramsey1.chem.uic.edu/chem232/page7/files/Chem%20232%20Lecture%2025.pdf)

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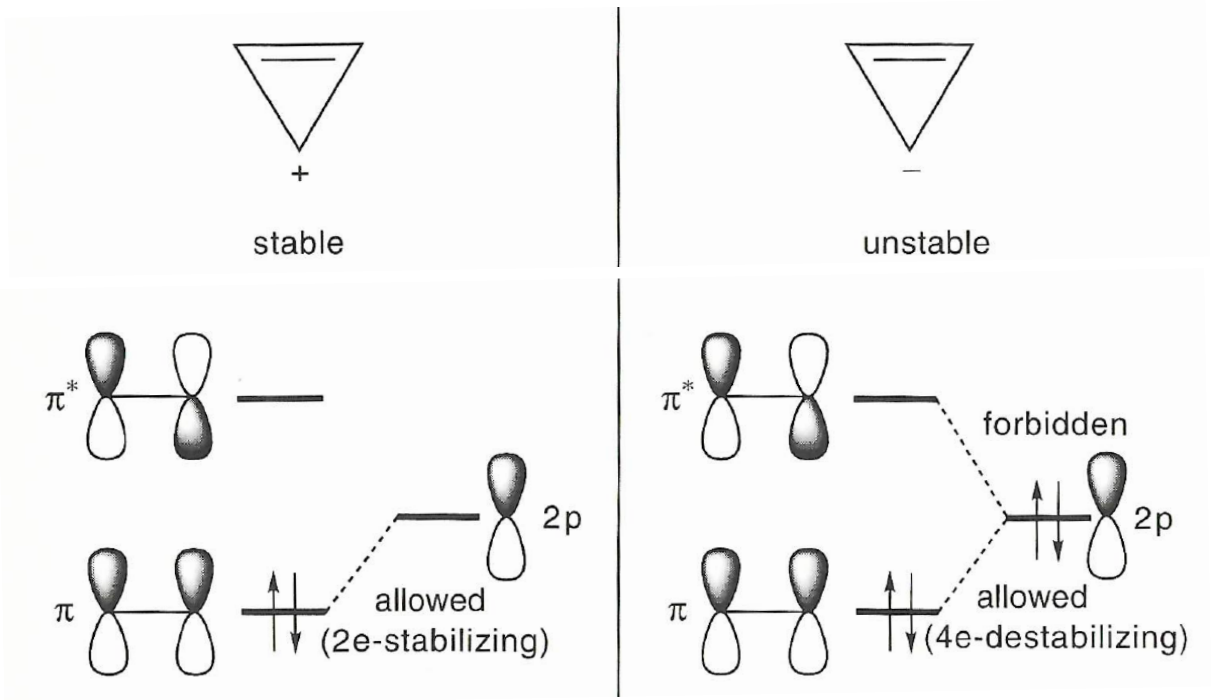
Symmetry Argument



(Image: Pross, *Theoretical & Physical Principles of Organic Reactivity*, Figure 2.26)